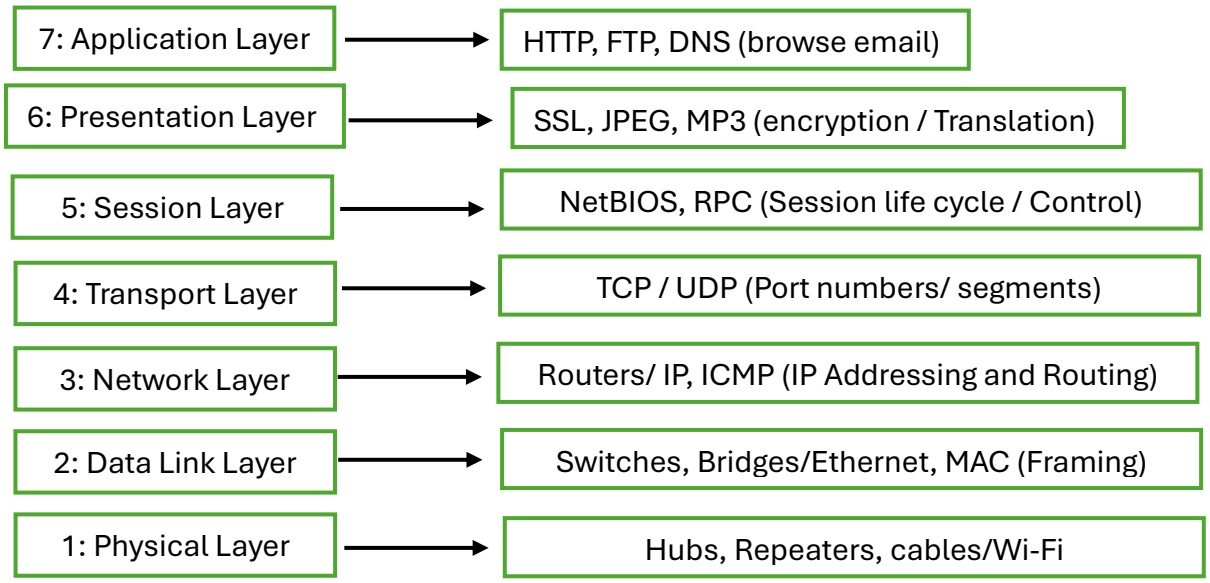
1. Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).



2. Match the following networking scenarios to the correct OSI layer:

a) Encrypting data before sending, **b) Routing packets across networks,**

c) Converting data to electrical signals, **d) Opening a web page in your browser,**

e) Detecting and correcting errors in frames.

Write your answers as 'Scenario: Layer Name (Layer Number)'.

a) Encrypting data before sending: Presentation Layer (Layer 6)

b) Routing packets across networks: Network Layer (Layer 3)

c) Converting data to electrical signals: Physical Layer (Layer 1)

d) Opening a web page in your browser: Application Layer (Layer 7)

e) Detecting and correcting errors in frames: Data Link Layer (Layer 2)

3. Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

When you use WhatsApp to send a simple text message, your data travels down the OSI (Open Systems Interconnection) model on your phone, across the internet, and back up the model on the recipient's phone.

Here is how three different OSI layers are directly involved in that process:

1. The Application Layer (Layer 7)

This layer interacts directly with the software you are using. When we type a message and hit send, This layer initiates the process.

- What it does: it handles the protocol that allows the app to communicate with network services.
- WhatsApp's Role: WhatsApp uses a specialized version of the XMPP (Extensible Messaging and Presence Protocol) or similar custom protocols to format your chat data, manage your "online" status, and trigger the delivery notification ticks.

2. The Presentation Layer (Layer 6)

This layer acts as the data translator. It ensures that the data sent by the application layer is formatted in a way that the receiving device can understand, while also handling security.

- What it does: It manages data formatting, data compression and encryption.
- WhatsApp's Role: his layer is critical for WhatsApp's famous end-to-end encryption (Signal Protocol). It takes your plain text message and scrambles it into unreadable ciphertext before it leaves your device. It also compresses the data so it uses less bandwidth.

3. Transport Layer (Layer 4)

This layer is responsible for getting the data reliably from your device to WhatsApp's server without errors.

- What it does: It breaks long messages into smaller pieces called segments, ensures they arrive in the correct order, and checks for data loss.
- WhatsApp's Role: WhatsApp primarily relies on the TCP (Transmission Control Protocol) for text messages. TCP ensures "reliable delivery," meaning if a segment of your text gets lost due to a spotty cellular connection, this layer requests a retransmission so your friend doesn't receive a broken or scrambled message.

4. Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

The problem likely stems from Layer 4 (Transport Layer) or Layer 7 (Application Layer), or involves

Layer 6 (Presentation Layer) issues like TLS/SSL failures

Because a ping uses ICMP (Internet Control Message Protocol) at Layer 3 (Network Layer), a successful ping proves that IP addressing, basic routing, and Layer 1 or 2 physical and data link connectivity between your device and the target server are working.

Layer 4 (Transport Layer — TCP/UDP)

Reason: A ping does not use TCP or UDP; it uses ICMP. The web server might be blocking or failing to respond to TCP port 80 (HTTP) or port 443 (HTTPS) due to a host-based firewall, a security group

restriction, or a down web service, even though the underlying IP address responds to ICMP echo requests.

Layer 7 (Application Layer):

Reason**:** The web application itself may have crashed, returned a server-side error (such as a 5xx code or misconfigured virtual host), or the DNS name mapped to a different service than the one you are pinging

Layer 6 (Presentation Layer):

Reason**:** SSL/TLS handshake failures expired or untrusted digital certificates, or cipher mismatches can prevent a browser from establishing an encrypted session after the TCP connection is made.

Proxy or Application-Level Filtering:

Misconfigured browser proxy settings, corporate web filters, or VPN tunnels can block web traffic while allowing raw IP ICMP pings to pass through.